MAT 3007 Homework # 6 Spring 2024

Due: Friday, April 19 at 12:00pm (noon)

Please note: You should submit your answers (scanned or typeset) in a PDF file. The problem set is due at 12:00pm (noon) on Friday, April 19th.

1. (30 pts) Consider the following functions:

$$\begin{aligned} a(x) &= 2x_1^2 - 30x_1 + 10 \\ b(x) &= -4x_2^2 + 20x_2 \\ c(x) &= \ln(x_1 + 1) + \ln(x_2 + 2) \\ d(x) &= 5x_1^2 + 3x_2^2 \\ e(x) &= -2x_1^2 + x_1x_2 - 4x_2^2 \\ f(x) &= x_1 + x_2 \end{aligned}$$

For each of the following, indicate whether the feasible region is a convex set, and classify each model as to whether it is a convex program:

(a) $\min\{a(x) - b(x) : f(x) \leq 10\}$

(b) $\max\{b(x) + c(x) : d(x) \leq 10; x \geq 0\}$

(c) $\max\{d(x) : c(x) \geq 4, e(x) \geq -5, f(x) \leq 10, x \geq 0\}$

(d) $\min\{d(x) : c(x) \geq 4, e(x) \geq -5, f(x) \leq 10, x \geq 0\}$

2. (30 pts) Is the following function convex?

$$f(x) = 5x_1 + 2x_2^2 + x_3^2 - 3x_3x_4 + 4x_4^2 + 2x_5^4 + x_5^2 + 3x_5x_6 + 6x_6^2 + 3x_6x_7 + x_7^2$$

Hint: Try to answer without building a Hessian that is any larger than need be.

3. (40 pts) Let $m \in \mathbb{R}^n$ and $r > 0$ be given and define the ball $C \coloneqq \{x \in \mathbb{R}^n : \|x-m\|_2^2 \leq r^2\}$, where $\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$ stands for the two-norm. We want to compute the projection of any point $x \in \mathbb{R}^n$ onto the ball, i.e., we want to solve the optimization problem:

$$\begin{aligned} \min_{y \in \mathbb{R}^n} \quad & \frac{1}{2}\|y - x\|^2 \\ \text{s.t.} \quad & \|y - m\|^2 \leq r^2. \end{aligned}$$

(a) Write down the KKT conditions for this projection problem.

(b) Show that the KKT conditions have a unique solution and calculate the corresponding KKT primal-dual pair explicitly.